Instrumental Variables

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Motivation

Three important threats to internal validity:

- Omitted variables (Unobserved, so cannot be included in the regression) – If not constant over time or states, panel data do not help
- Measurement error (X is measured with error)
- Simultaneity (X causes Y, Y causes X)

All of them imply that $E[u|X] \neq 0$, and OLS estimator is biased and inconsistent.

Instrumental variables (IV) regression can eliminate bias from these three sources.

What Is IV?

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

- Given that X is correlated with u, we can break X into two parts: a part that might be correlated with u, and a part that is not. By isolating the part that is not correlated with u, it is possible to estimate β_1
- This can be done by an instrumental variable (IV), denoted as Z, which is uncorrelated with u
- The IV Z detects movements in X that are uncorrelated with u, and thus estimate β_1

Two Conditions: Validity of IV

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$

For an instrumental variable (an "instrument") Z to be valid, it must satisfy two conditions:

1. Relevance: $corr(Z, X) \neq 0$
2. Exogeneity: $corr(Z, u) = 0$

Alternatively, you may explain: Z affects Y only through X.

- IV is not simply a proxy for the causal variable of interest
- IV is not an omitted variable

Suppose for now that you have such a Z (we'll discuss how to find instrumental variables later). How can you use Z to estimate β_1?

Two Stage Least Squares (2SLS)

As it sounds, 2SLS has two stages – two regressions:

1. First, isolate the part of X that is uncorrelated with u: regress X on Z using OLS

$$X_i = \pi_0 + \pi_1 Z_i + v_i \tag{1}$$

   - Because Z_i is uncorrelated with u_i, $\pi_0 + \pi_1 Z_i$ is uncorrelated with u_i. We don't know π_0 or π_1 (population parameters) but we have estimated them, so ...
   - Compute the predicted values of X_i, $\hat{X}_i$, where $\hat{X}_i = \hat{\pi}_0 + \hat{\pi}_1 Z_i, i = 1, \ldots, n$.

Two Stage Least Squares (2SLS)

2. Replace X_i by $\hat{X}_i$ in the regression of interest: regress Y on $\hat{X}_i$ using OLS:

$$Y_i = \beta_0 + \beta_1 \hat{X}_i + u_i \qquad (2)$$

- Because $\hat{X}_i$ is uncorrelated with u_i in large samples, so the first least squares assumption holds
- Thus, β_1 can be estimated by OLS using regression (2)
- This argument relies on large samples (so π_0 and π_1 are well estimated using regression (1))
- This the resulting estimator is called the "Two Stage Least Squares" (2SLS) estimator. It is also called IV estimator

Two Stage Least Squares (2SLS)

$$\beta_1 = \frac{cov(Z, Y)}{cov(Z, X)}$$

- β_1 is equal to the ratio of two covariances
- Thus, we can estimate β_1 by replacing these population covariances with sample covariances – this is the method of moment estimator
- The resulting estimator is identical to $\hat{\beta}_{1,IV}$, so the IV estimator can be explained as the method of moment estimator

Note: If $Z = X$, the IV estimator becomes the OLS estimator!

General IV Regression

- So far, we have considered IV regression with a single endogenous regressor (X) and a single instrument (Z)
- We can extend this to:
  - multiple endogenous regressors ($X_1, \ldots, X_k$)
  - multiple included exogenous variables ($W_1, \ldots, W_r$) – These need to be included for the usual OV reason
  - multiple instrumental variables ($Z_1, \ldots, Z_m$) – More relevant instruments can produce a smaller variance of 2SLS

General IV Regression

The general IV regression model:

$$Y_i = \beta_0 + \beta_1 X_{1i} + \cdots + \beta_k X_{ki} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r} W_{ri} + u_i \, .$$

- Y_i is the dependent variable
- $X_{1i}, \ldots, X_{ki}$ are the endogenous regressors (potentially correlated with u_i)
- $W_{1i}, \ldots, W_{ri}$ are the control variables or included exogenous regressors/varibles (uncorrelated with u_i)
- $\beta_0, \beta_1, \ldots, \beta_{k+r}$ are the unknown regression coefficients
- $Z_{1i}, \ldots, Z_{mi}$ are the m instrumental variables (the excluded exogenous variables) ($m \geq k$ (why this requirement?))

Identification

- In general, a parameter is said to be identified if different values of the parameter produce different distributions of the data
- In IV regression, whether the coefficients are identified depends on the relation between the number of instruments (m) and the number of endogenous regressors (k)
- Intuitively, if there are fewer instruments than endogenous regressors, we can't estimate $\beta_1, \ldots, \beta_k$. For example, $k = 1$ but $m = 0$ (no instruments)!

Identification

The coefficients $\beta_1, \ldots, \beta_k$ are said to be

- exactly identified if $m = k$. There are just enough instruments to estimate $\beta_1, \ldots, \beta_k$
- overidentified if $m > k$. There are more than enough instruments to estimate $\beta_1, \ldots, \beta_k$
- underidentified if $m < k$. There are too few instruments to estimate $\beta_1, \ldots, \beta_k$. If so, you need to get more instruments!

Two Stage Least Squares (2SLS)

1. First, isolate the part of X_j that is uncorrelated with u:regress X_j on $Z_1, \ldots, Z_m$ and $W_1, \ldots, W_r$ using OLS

$$X_{ji} = \pi_{j,0} + \pi_{j,1}Z_{1i} + \cdots + \pi_{j,m}Z_{mi} + \pi_{j,m+1}W_{1i} + \cdots + \pi_{j,m+r}W_{ri} + v_{ji} \text{ for } j = 1, \ldots, k \quad (3)$$

   - Because $Z_{1i}, \ldots, Z_{mi}$ and $W_{1i}, \ldots, W_{ri}$ are uncorrelated with u_i, $\pi_{j,0} + \pi_{j,1}Z_{1i} + \cdots + \pi_{j,m}Z_{mi} + \pi_{j,m+1}W_{1i} + \cdots + \pi_{j,m+r}W_{ri}$ is uncorrelated with u_i. We don't know $\pi_{j,0}, \pi_{j,1}, \ldots, \pi_{j,m+r}$ but we have estimated them, so ...
   - Compute the predicted values of X_{ji}, $\hat{X}_{ji}$, where $\hat{X}_{ji} = \hat{\pi}_{j,0} + \hat{\pi}_{j,1}Z_{1i} + \cdots + \hat{\pi}_{j,m}Z_{mi} + \hat{\pi}_{j,m+1}W_{1i} + \cdots + \hat{\pi}_{j,m+r}W_{ri}, i = 1, \ldots, n.$

Why W's are included? We want to test whether instruments can predict X's after controlling W's

Two Stage Least Squares (2SLS)

2. Replace X_{ji} by $\hat{X}_{ji}$ in the regression of interest: regress Y on $\hat{X}_{1i}, \ldots, \hat{X}_{ki}$ and $W_{1i}, \ldots, W_{ri}$ using OLS:

$$Y_i = \beta_0 + \beta_1 \hat{X}_{1i} + \cdots + \beta_k \hat{X}_{ki} + \beta_{k+1} W_{1i} + \cdots + \beta_{k+r} W_{ri} + u_i \quad (4)$$

   - Because $\hat{X}_{ji}$ is uncorrelated with u_i in large samples, so the first least squares assumption holds
   - Thus, $\beta_0, \beta_1, \ldots, \beta_{k+r}$ can be estimated by OLS using regression (4)
   - This argument relies on large samples (so $\pi_{j,0}, \pi_{j,1}, \ldots, \pi_{j,m+r}$ are well estimated using regression (3))
   - This the resulting estimator is called the "Two Stage Least Squares" (2SLS) estimator. We do not call it as IV estimator unless $k = m$

Two Stage Least Squares (2SLS)

- Solving two multiple linear regressions, you may derive the formula for the 2SLS estimator – It is more complicated than in the case of single X and Z. It is covered in graduate econometric course, but not here!
- The connection with the method of moment can be established as well. Will be covered in graduate econometric course

You may use Stata to get the result easily. In Stata, you may enter

ivreg Y (X1 X2 =Z1 Z2 Z3) W1 W2 W3,r
or ivreg Y W1 W2 W3 (X1 X2 =Z1 Z2 Z3), r

Exogeneity & Relevance Condition

Question: What is the relevance condition for $k > 1$? It can be defined as follows:

- There is no perfect multicolinearity for $\{\pi_{j,0} + \pi_{j,1}Z_{1i} + \cdots + \pi_{j,m}Z_{mi} + \pi_{j,m+1}W_{1i} + \cdots + \pi_{j,m+r}W_{ri}\}_{j=1}^{k}$ and $\{W_{ji}\}_{j=1}^{r}$
- When $k = 1$, the vector $(\pi_{1,1}, \ldots, \pi_{1,m})$ is not equal to zero
- When $k = 1$, $r = 0$ and $m = 1$, the correlation between IV and regressor is nonzero

Exogeneity condition: $Cov(Z_{1i}, u_i) = 0, \ldots, Cov(Z_{mi}, u_i) = 0$

Check Instrument Validity

Focus on a single endogenous regressor. Recall the two requirements for valid instrument:

- Relevance: at least one instrument must enter the population counterpart of the first stage regression. That is, the vector $(\pi_1, \ldots, \pi_m)$ is not equal to zero
- Exogeneity: all the instruments must be uncorrelated with the error term. That is, $Cov(Z_{1i}, u_i) = 0, \ldots, Cov(Z_{mi}, u_i) = 0$

What happens if one of these two requirements isn't satisfied? How do you check? What do you do? If you have multiple instruments, which should you use?

Check Instrument Validity

Checking instrument relevance. The model with a single included endogenous regressor:

$$Y_i = \beta_0 + \beta_1 X_i + \beta_2 W_{1i} + \cdots + \beta_{1+r} W_{ri} + u_i \, .$$

The first stage regression:

$$X_i = \pi_0 + \pi_1 Z_{1i} + \cdots + \pi_m Z_{mi} + \pi_{m+1} W_{1i} + \cdots + \pi_{m+r} W_{ri} + v_i \, .$$

- The IVs are relevant if one of $\pi_1, \ldots, \pi_m$ is nonzero. Sometimes, we say the instruments are strong
- If $\pi_1, \ldots, \pi_m$ are all zero, the IVs are said to be irrelevant
- The IVs are said to be weak if all $\pi_1, \ldots, \pi_m$ are nearly zero (how close?)

Check Instrument Validity

How to test for relevance condition? It amounts to testing for

$$H_0 : \pi_1 = \ldots = \pi_m = 0 .$$

- It can be done by F test in the first stage regression
- When $m = 1$, we reject H_0 if $F > 3.84$ at 5% level. We may conclude that the relevance condition is likely to be true
- However, this does not imply that the sampling distribution of the 2SLS/IV estimator is normal for large n
- In applications, people usually require that $F > 10$ for $m = 1$. If $F < 10$, then the IV is weak. This is the definition of weak IV
- The definition of weak IVs for multiple instruments is given by Stock and Yogo (2005)

Check Instrument Validity

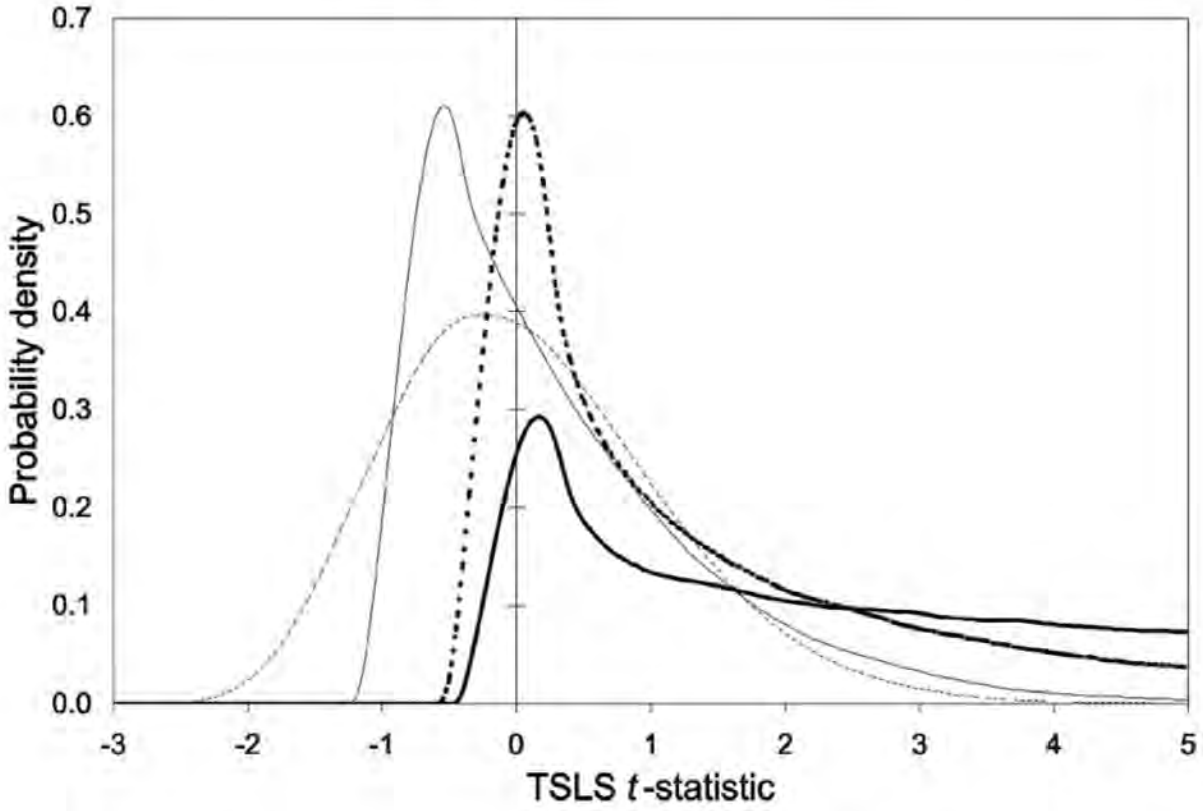
If instruments are weak, the sampling distribution of $\hat{\beta}_{1,IV}$ is not normal, even with n large. Intuition: consider the simplest case

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$
$$X_i = \pi_0 + \pi_1 Z_i + v_i \, .$$

- The IV estimator $\hat{\beta}_{1,IV} = S_{YZ}/S_{XZ} = \frac{S_{YZ}/S_Z^2}{S_{XZ}/S_Z^2} = \frac{S_{YZ}/S_Z^2}{\hat{\pi}_1}$
- In the first stage regression, $F = (\hat{\pi}_1/SE(\hat{\pi}_1))^2$. The sampling distribution of $\hat{\beta}_{1,IV}$ depends on the value of F
- With weak instruments, F is not large and then the sampling distribution of $\hat{\beta}_{1,IV}$ is not well approximated by its large n normal approximation

Check Instrument Validity

An example: the sampling distribution of $\hat{\beta}_{1,IV}$ with weak IV



Dark line: irrelevant IV; dashed light line: strong IV

Check Instrument Validity

Why does our trusty normal approximation fail us? Mathematical reason: consider $\pi_1 = 0$ (irrelevant IV)

$$\hat{\beta}_{1,IV} - \beta_1 \approx \frac{n^{-1/2} \sum_{i=1}^{n} (Z_i - \mu_Z) u_i}{n^{-1/2} \sum_{i=1}^{n} (Z_i - \mu_Z)(X_i - \mu_X)} .$$

- Both the numerator and denominator are asymptotically normal, so the sampling distribution of $\hat{\beta}_{1,IV} - \beta_1$ is better approximated by a ratio of two normals (which is related to Cauchy distribution)
- If instruments are weak, the usual methods of inference are unreliable – potentially very unreliable
- Rule of thumb: $F > 10$. If $F < 10$, you may have weak IV issue ($\hat{\beta}_{1,IV}$ is biased and its associated t test is misleading)

Check Instrument Validity

What to do if you have weak instruments?

- Get better instruments (often easier said than done!)
- If you have many instruments, some are probably weaker than others and it is a good idea to drop the weaker ones (dropping an irrelevant instrument will increase the first stage F)
- If you only have a few instruments, and all are weak, then you need to conduct inference using other methods (rather than using t or F tests based on the 2SLS estimator)

Check Instrument Validity

Inference with weak instruments

- Anderson-Rubin test: $H_0 : \beta_1 = \beta_{1,0}$ v.s. $H_1 : \beta_1 \neq \beta_{1,0}$
  - Compute $\epsilon_i = Y_i - X_i\beta_{1,0}$
  - Regress ϵ_i on $W_{1i}, \ldots, W_{ri}, Z_{1i}, \ldots, Z_{mi}$
  - The AR test is the F test on the joint significance of the coefficients of $Z_{1i}, \ldots, Z_{mi}$
  - The method is robust to weak instruments or irrelevant instrument
- Alternative method: Moreira's conditional likelihood ratio test (this method is more powerful when there are many instruments)

Check Instrument Validity

Checking instrument exogeneity. Recall that all the instruments must be uncorrelated with the error term. That is, $Cov(Z_{1i}, u_i) = 0, \ldots, Cov(Z_{mi}, u_i) = 0$.

Consider the simplest case

$$Y_i = \beta_0 + \beta_1 X_i + u_i$$
$$X_i = \pi_0 + \pi_1 Z_i + v_i \, .$$

The natural option is to check whether the sample covariance between Z_i and $\hat{u}_i = Y_i - \hat{\beta}_{0,IV} - \hat{\beta}_{1,IV} X_i$ is far away from zero.

- What is the problem with this method? What is $S_{Z\hat{u}} = \widehat{Cov}(Z, \hat{u}) = n^{-1} \sum_{i=1}^{n} (Z_i - \bar{Z})(\hat{u}_i - \bar{\hat{u}})$?

Check Instrument Validity

Testing for instrument exogeneity can be done in the case of overidentification. Consider the simple case

$$Y_i = \beta_0 + \beta_1 X_i + u_i \, .$$

Suppose there are two valid instruments: Z_{1i}, Z_{2i}.

- Then you could compute two separate 2SLS estimates
- Intuitively, if these two 2SLS estimates are very different from each other, then something must be wrong: one or the other (or both) of the instruments must be invalid.
- The J-test of overidentifying restrictions makes this comparison in a statistically precise way.
- This can only be done if $m > k$ (overidentified). The J-test is also known as overidentification test

Check Instrument Validity

The J-test is the Anderson-Rubin test, using the 2SLS estimator instead of the hypothesized value $\beta_{1,0}$. The recipe:

- Obtain the 2SLS estimator using all m instruments; compute the predicted values $\hat{Y}_i$ using the actual X's
- Compute the residual $\hat{u}_i = Y - \hat{Y}_i$
- Regress $\hat{u}_i$ on $W_{1i}, \ldots, W_{ri}, Z_{1i}, \ldots, Z_{mi}$
- Compute the F-statistic testing the hypothesis that the coefficients on $Z_{1i}, \ldots, Z_{mi}$ are all zero
- The J-statistic is $J = mF$

Check Instrument Validity

Distribution of the J-statistic

- Under the null hypothesis that all the instruments are exogeneous, J has a chi-squared distribution with $m-k$ degrees of freedom
- If $m = k$, $J = 0$ (does this make sense?)
- If some instruments are exogenous and others are endogenous, the J statistic will be large, and the null hypothesis that all the instruments are exogeneous will be rejected

Check Instrument Validity

We mainly focus on a single X.

Relevance

- At least one instrument must enter the population counterpart of the first stage regression
- If instruments are weak, then the 2SLS estimator is biased and the t statistic has non-normal distribution
- To check for weak instruments with a single endogenous regressor, check the first-stage F
  - If $F > 10$, instruments are strong – use 2SLS estimator
  - If $F > 10$, instruments are weak – take some action

Check Instrument Validity

Exogeneity

- All the instruments must be uncorrelated with the error term: $Cov(Z_{1i}, u_i) = 0, \ldots, Cov(Z_{mi}, u_i) = 0$.
- We can partially test for exogeneity: if $m > 1$, we can test the null hypothesis that all the instruments are exogenous, against the alternative that as many as $m-1$ are endogenous (correlated with u)
- The test is the J-test, which is constructed using the 2SLS residuals
- If the J-test rejects, then at least some of your instruments are endogenous – so you must make a difficult decision and jettison some (or all) of your instruments

Application: Demand for Cigarettes

Why are we interested in knowing the elasticity of demand for cigarettes?

- Theory of optimal taxation. The optimal tax rate is inversely related to the price elasticity: the greater the elasticity, the less quantity is affected by a given percentage tax, so the smaller is the change in consumption and deadweight loss
- Externalities of smoking – role for government intervention to discourage smoking
  - health effects of second-hand smoke? (non-monetary)
  - monetary externalities

Application: Demand for Cigarettes

Data set

- Annual cigarette consumption, average prices paid by end consumer (including tax), personal income, and tax rates (cigarette-specific and general statewide sales tax rates)
- 48 continental US states, 1985-1995 (panel data)

Estimation strategy

- We need to use IV estimation methods to handle the simultaneous causality bias that arises from the interaction of supply and demand
- Control variables $W's$ = State binary indicators, which control for unobserved state-level characteristics that affect the demand for cigarettes and the tax rate, as long as those characteristics don't vary over time

Application: Demand for Cigarettes

$$\ln(Q_{it}^{cigarettes}) = \alpha_i + \beta_1 \ln(P_{it}^{cigarettes}) + \beta_2 \ln(Income_{it}) + u_{it}$$
$$i = 1, \ldots, 48, t = 1985, \ldots, 1995$$

- $Corr(\ln(P_{it}^{cigarettes}), u_{it})$ is plausibly nonzero because of supply and demand interactions
- α_i reflects unobserved omitted factors which vary across states but not over time, e.g., attitude towards smoking

Estimation method

- Use panel data regression methods to eliminate α_i
- Use 2SLS estimator to handle simultaneous causality bias
- Two possible instruments: general sales tax and cigarette-specific tax

Application: Demand for Cigarettes

First stage regression:

```
reg lravgprs rtaxso if year==1995, r

Linear regression                               Number of obs =      48
                                                F(  1,    46) =   40.39
                                                Prob > F      =  0.0000
                                                R-squared     =  0.4710
                                                Root MSE      =  .09394

------------------------------------------------------------------------------
             |               Robust
    lravgprs |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
      rtaxso |   .0307289   .0048354     6.35   0.000     .0209956    .0404621
       _cons |   4.616546   .0289177   159.64   0.000     4.558338    4.674755
------------------------------------------------------------------------------
. predict lravphat
```

$$\widehat{\ln(P^{cigarettes})} = \underset{(0.04)}{4.63} + \underset{(0.004)}{0.031} \times SaleTax, n = 48$$

Application: Demand for Cigarettes

Second stage regression:

```
reg lpackpc lravphat if year==1995, r

Linear regression                               Number of obs =      48
                                                F(  1,    46) =   10.54
                                                Prob > F      =  0.0022
                                                R-squared     =  0.1525
                                                Root MSE      =  .22645

------------------------------------------------------------------------------
             |               Robust
     lpackpc |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    lravphat |  -1.083586   .3336949    -3.25   0.002    -1.755279   -.4118932
       _cons |   9.719875   1.597119     6.09   0.000     6.505042    12.93471
------------------------------------------------------------------------------
```

$$\widehat{\ln(Q^{cigarettes})} = \underset{(1.60)}{9.72} - \underset{(0.33)}{1.08} \times \widehat{\ln(P^{cigarettes})}, n = 48$$

- These coefficients are the IV estimates
- The standard errors are wrong because they ignore the fact that the first stage was estimated

Application: Demand for Cigarettes

```
. ivreg ltpackpc (ltavgprs = dtrtaxso dtrtax) ltperinc, r

Instrumental variables (2SLS) regression          Number of obs =      48
                                                  Wald chi2(2)  =   45.44
                                                  Prob > chi2   =  0.0000
                                                  R-squared     =  0.5466
                                                  Root MSE      =  .08836

------------------------------------------------------------------------------
             |               Robust
    ltpackpc |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    ltavgprs |  -1.202403   .1906896    -6.31   0.000    -1.576148   -.8286588
    ltperinc |   .4620299   .2995177     1.54   0.123    -.1250139    1.049074
       _cons |   .3665388   .1180414     3.11   0.002     .1351819    .5978957
------------------------------------------------------------------------------
Instrumented:  ltavgprs
Instruments:   ltperinc dtrtax dtrtaxso
------------------------------------------------------------------------------
```

- Use both general sales tax and cigarette-specific tax as instruments

Application: Demand for Cigarettes

```
. reg ltavgprs dtrtaxso dtrtax ltperinc

      Source |       SS       df       MS              Number of obs =      48
-------------+------------------------------           F(  3,    44) =   51.36
       Model |  .289359873     3  .096453291            Prob > F      =  0.0000
    Residual |  .082627329    44  .001877894            R-squared     =  0.7779
-------------+------------------------------           Adj R-squared =  0.7627
       Total |  .371987202    47  .007914621            Root MSE      =  .04333

------------------------------------------------------------------------------
    ltavgprs |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    dtrtaxso |    .013457   .0030498     4.41   0.000     .0073106    .0196033
      dtrtax |   .0075734   .0010488     7.22   0.000     .0054597     .009687
    ltperinc |  -.0289943   .1474923    -0.20   0.845    -.3262455    .2682568
       _cons |   .4919733   .0220923    22.27   0.000     .4474492    .5364973
------------------------------------------------------------------------------

.  test dtrtaxso dtrtax

 ( 1)  dtrtaxso = 0
 ( 2)  dtrtax = 0
       F(  2,    44) =   75.65
            Prob > F =    0.0000
```

- 75.65 > 10, so instruments aren't weak
- With $m > k$, we can test instrument exogeneity

Application: Demand for Cigarettes

```
. predict e, resid;          **Computes predicted values for most recently**
                          **estimated regression (the previous TSLS regression)**
. reg e dtrtaxso dtrtax ltperinc  **Regress e on Z's and W's**

      Source |       SS       df       MS              Number of obs =      48
-------------+------------------------------           F(  3,    44) =    1.64
       Model |  .037769176     3  .012589725            Prob > F      =  0.1929
    Residual |  .336952289    44  .007658007            R-squared     =  0.1008
-------------+------------------------------           Adj R-squared =  0.0395
       Total |  .374721465    47  .007972797            Root MSE      =  .08751

------------------------------------------------------------------------------
           e |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
    dtrtaxso |   .0127669   .0061587     2.07   0.044      .000355    .0251789
      dtrtax |  -.0038077   .0021179    -1.80   0.079     -.008076    .0004607
    ltperinc |  -.0934062   .2978459    -0.31   0.755    -.6936752    .5068627
       _cons |    .002939   .0446131     0.07   0.948    -.0869728    .0928509
------------------------------------------------------------------------------

. test dtrtaxso dtrtax
 ( 1)  drtaxso = 0
 ( 2)  drtax = 0

       F(  2,    44) =    2.47
            Prob > F =    0.0966 ** WARNING { this uses the wrong d.f. **
```

Application: Demand for Cigarettes

- $J = mF = 2 * 2.47 = 4.93$, compare it with chi-squared with d.f. = 1; the 5% critical value is 3.84, so we reject at 5% significance level
- The correct degrees of freedom for the J-statistic is $m - k$
- In Stata:

```
. dis "J-stat = " r(df)*r(F) "  p-value = "  chiprob(r(df)-1,r(df)*r(F));
J-stat = 4.9319853  p-value = .02636401
```

- Now what?

Application: Demand for Cigarettes

TABLE 12.1 Two Stage Least Squares Estimates of the Demand for Cigarettes Using Panel Data for 48 U.S. States

Dependent variable: $\ln(Q_{i,1995}^{cigarettes}) - \ln(Q_{i,1985}^{cigarettes})$

Regressor	(1)	(2)	(3)
$\ln(P_{i,1995}^{cigarettes}) - \ln(P_{i,1985}^{cigarettes})$	−0.94** (0.21)	−1.34** (0.23)	−1.20** (0.20)
$\ln(Inc_{i,1995}) - \ln(Inc_{i,1985})$	0.53 (0.34)	0.43 (0.30)	0.46 (0.31)
Intercept	−0.12 (0.07)	−0.02 (0.07)	−0.05 (0.06)
Instrumental variable(s)	Sales tax	Cigarette-specific tax	Both sales tax and cigarette-specific tax
First-stage *F*-statistic	33.70	107.20	88.60
Overidentifying restrictions *J*-test and *p*-value	—	—	4.93 (0.026)

These regressions were estimated using data for 48 U.S. states (48 observations on the 10-year differences). The data are described in Appendix 12.1. The *J*-test of overidentifying restrictions is described in Key Concept 12.6 (its *p*-value is given in parentheses), and the first-stage *F*-statistic is described in Key Concept 12.5. Individual coefficients are statistically significant at the *5% level or **1% significance level.

Application: Demand for Cigarettes

How should we interpret the J-test rejection?

- J-test rejects the null hypothesis that both the instruments are exogenous
- This means that either general sales tax is endogenous, or cigarette-specific tax is endogenous, or both!
- The J-test doesn't tell us which! You must exercise judgment . . .
- Why might cigarette-specific tax be endogenous?
  - Political forces: history of smoking or lots of smokers – political pressure for low cigarette taxes
  - If so, cigarette-specific tax is endogenous
- This reasoning doesn't apply to general sales tax
- Use just one instrument, the general sales tax

Application: Demand for Cigarettes

Summary:

- Use the estimated elasticity based on 2SLS with the general sales tax as the only instrument: Elasticity = −.94, SE = .21
- This elasticity is surprisingly large (not inelastic) – a 1% increase in prices reduces cigarette sales by nearly 1%. This is much more elastic than conventional wisdom in the health economics literature
- This is a long-run (ten-year change) elasticity. *What would you expect a short-run (one-year change) elasticity to be – more or less elastic?*